

1 Verilog Implementation

```
1 module top_tb;
2
3     reg clk, rst;
4     reg code1_in;
5     reg [3:0] code2_in;
6     reg [2:0] code3_in;
7     reg [7:0] code4_in;
8     reg [1:0] code5_in;
9
10    wire phase1_done, phase1_fail;
11    wire phase2_done, phase2_fail;
12    wire phase3_done, phase3_fail;
13    wire phase4_done, phase4_fail;
14    wire phase5_done, phase5_fail;
15    wire all_done;
16
17    top dut (
18        .clk(clk),
19        .rst(rst),
20        .code1_in(code1_in),
21        .code2_in(code2_in),
22        .code3_in(code3_in),
23        .code4_in(code4_in),
24        .code5_in(code5_in),
25        .phase1_done(phase1_done),
26        .phase1_fail(phase1_fail),
27        .phase2_done(phase2_done),
28        .phase2_fail(phase2_fail),
29        .phase3_done(phase3_done),
30        .phase3_fail(phase3_fail),
31        .phase4_done(phase4_done),
32        .phase4_fail(phase4_fail),
33        .phase5_done(phase5_done),
34        .phase5_fail(phase5_fail),
35        .all_done(all_done)
36    );
37
38    initial begin
39        clk = 0;
40    end
41
42    always #5 clk = ~clk;
43
44    initial begin
45        $monitor("Time=%0t, rst=%b, code1_in=%b, phase1_done=%b,
46            phase1_fail=%b, code2_in=%b, phase2_done=%b, phase2_fail=%b,
47            code3_in=%b, phase3_done=%b, phase3_fail=%b, code4_in=%b,
48            phase4_done=%b, phase4_fail=%b, code5_in=%b, phase5_done
49            =%b, phase5_fail=%b, all_done=%b" ,
```

```

46         $time, rst,
47         code1_in, phase1_done, phase1_fail,
48         code2_in, phase2_done, phase2_fail,
49         code3_in, phase3_done, phase3_fail,
50         code4_in, phase4_done, phase4_fail,
51         code5_in, phase5_done, phase5_fail,
52         all_done);
53
54     rst = 1;
55     #10;
56     rst = 0;
57     #10;
58
59     code1_in = 1; #10;
60     code1_in = 0; #10;
61     code1_in = 1; #10;
62     code1_in = 1; #10;
63
64     #20;
65     rst = 1; #10;
66     rst = 0; #10;
67     code2_in = 4'b1101; #40;
68
69     rst = 1; #20;
70     rst = 0;
71     #10 code3_in = 3'b000;
72     #10 code3_in = 3'b001;
73     #10 code3_in = 3'b010;
74     #10 code3_in = 3'b011;
75     #10 code3_in = 3'b100;
76
77     #20;
78     rst = 1; #10;
79     rst = 0; #10;
80     code4_in = 8'b10101010; #10;
81     code4_in = 8'b11001100; #10;
82     code4_in = 8'b11110000; #10;
83
84     #20;
85     rst = 1; #20;
86     rst = 0;
87
88     #10 code5_in = 2'b01;
89     #10 code5_in = 2'b10;
90     #10 code5_in = 2'b11;
91
92     #20;
93     $finish;
94 end
95 initial begin
96     $dumpfile("dump.vcd");

```

```
97     $dumpvars(0, top_tb);
98 end
99
100
101 endmodule
```